

Arjun Giriraj Day 11

1. What is Kali Linux?
Kali linux is a debian based operating system used for cybersecurity, security testing, penetration testing, digital forensics, security research, network security.
2. Why is Kali Linux used in cybersecurity?
Kali linux contains powerful tools which are used by security tester, penetration tester, digital forensics investigators, etc.
3. What is a virtual machine?
Virtual machine runs another operating system inside the virtual controlled environment. It is all done in host operating system.
4. What is a hypervisor?
It is a software that creates or manage virtual machines.
5. What is VirtualBox?
It is the software that lets you create virtual machines.
6. Explain Host OS and Guest OS.
Host OS is the main OS that is installed on your physical machine; Guest OS is run inside the virtual machine that lets you use another OS beside your Host OS.
7. Why is using Kali inside a VM useful for beginners?
Because it lets you learn the kali linux without replacing the main os.
8. What is a VM snapshot?
It saves the VM state so you can return at the exact state you left your virtual machine.
9. Why should students create snapshots?
To use the virtual machine at the exact state student left it.
10. What does whoami display?
whoami displays the current user of the system.
11. What does hostname display?
hostname returns the computer's hostname.
12. What does uname -a display?
It returns the system information including kernel name, hostname, kernel version, etc.
13. What does /etc/os-release contain?
It contains the operating system identification data.
14. What does ip addr show?
Shows the computer's IP addresses, and network configuration.

15. What does ip route show?
It shows the routing information, and the default route.
16. What is sudo?
It gives user the privilege of root.
17. Why should elevated privileges be used carefully?
Because elevated privileges are the root privilege.
18. What does sudo apt update do?
It refreshes the local package repository information.
19. Why should Kali Linux tools only be used on authorized systems?
Kali linux tools used by penetration tester, offensive security engineer.
20. Explain the difference between ethical security testing and unauthorized access.
Ethical security testing is all done in authorized environment with permission from testing site; Unauthorized testing done using by unauthorized access of environment which is generally illegal.

SCREENSHOTS (from your own lab):

Kali desktop · whoami · hostname · uname -a · cat /etc/os-release · ip addr ·
jayvik-cyber-lab directory · VirtualBox snapshot (Day11-Clean-Kali)

```
venom@VENOM:~$ whoami
venom
venom@VENOM:~$ uname -a
Linux VENOM 7.1.5-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.1.5-1kali1 (2026-07-29) x86_64 GNU/Linux
venom@VENOM:~$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.3"
VERSION="2026.3"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```

```
venom@VENOM:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 10:6f:d9:7a:da:a9 brd ff:ff:ff:ff:ff:ff
    inet 10.245.202.85/24 brd 10.245.202.255 scope global dynamic noprefixroute wlan0
        valid_lft 2397sec preferred_lft 2397sec
    inet6 2401:4900:bcb1:9313:c018:3cc8:ad45:ef1a/64 scope global dynamic noprefixroute
        valid_lft 6744sec preferred_lft 6744sec
    inet6 fe80::83a4:13e:661d:1ec/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

I use Kali Linux as my main operating system, so I have not set it on a virtual machine.
But I have good knowledge in virtual machine setup also.